

THE COMMON EAVE

BEIJING AI URBAN TEST + PUBLIC-VALUE VERIFICATION CORRIDOR

FOUR BEIJING DUTIES · TWO ZONES AND ONE BELT · SEVEN PUBLIC-VALUE CHECKS

AI BACKSTAGE; PUBLIC LIFE FRONTSTAGE

01 EVIDENCE BEFORE FORM

Official facts / public observations / reproducible derivations / concepts remain distinct



Official scope + policy

43.6 km² / 11.4 km² / 368.4 ha



Public observation

ESA 2021 · 1248 cells



Derived correction

20 units · topology safe



Concept + stop rules

5 seams · 2 flagships

WHAT DID THE DATA ACTUALLY CHANGE?

- Several former green-band units show 62%-89% built signal → no belt-wide conversion
- Key-area vegetation signals 39.84% / 25.97% / 21.76% → three renewal regimes
- Dazhongsi local 3,190-person/h stress case → ground-first, exit-specific, load-shedding

02 THE PLAN IN ONE SENTENCE

Begin with retention · Shape through seams · Govern AI through reversibility

RETAIN

Innovation is not synonymous
with new construction

SEAM

Five real civic problems
organise 11.4 sq km

REVOKE

Named human ownership,
fallback, and stop before
scaling

AI does not decide the city; it exposes contradictions, replays evidence, and records who decided.

03 EVIDENCE-CORRECTED OVERALL PLAN

Retained clusters + narrow continuity + three built renewal programmes



- R&D/open
- retained cluster
- mixed service
- talent/public

Boundary remains provisional.
Retention clusters use 2021 public
land-cover signals; trees, rights, and
change require field checks.

04 FIVE SEAMS, NOT FIVE SLOGANS

Each seam: real problem → minimum civic move → professional gate → stop

Dazhongsi station-ground seam

Rail, bus, cycling, walking, project frontages, and delivery compete at peak arrival.

Build continuous ground, exit-specific routes, and timed logistics first; underground links remain a specialist option.

North Third Ring-Sidaokou safe-wait seam

Public-review records raise interchange classification and transfer-crossing questions; a concept line cannot claim resolution.

Start with sheltered waiting, crossing audits, non-digital wayfinding, and staffed help.

Zhichun-Jingbao heritage-everyday seam

Phase I is an existing base, while the Jingbao-Zhichun grade relationship still needs study.

Retain park operations; test reversible foyers, weather shelter, and everyday services in short segments.

Qinghuayuan-Wuda shared-learning seam

Campus, community, and heritage park are close, but civic learning, care waiting, and night interfaces remain weak.

Shared Eave combines free learning, repair, staffed help, care waiting, and open classes.

Qinghe-North Fifth Ring river-park seam

The north has a stronger vegetation signal; a uniform tech image could erase ecology and daily use.

Retain clusters; add a benchmark garden, stormwater observation, and repair-oriented tests.

05 THREE AREAS, THREE URBAN REGIMES

One identity does not mean one image; difference follows evidence, arrival, and civic problems

Zhongzhiyuan | open-source production landscape



2021 vegetation signal

Highest vegetation signal of the three. Retain wooded continuity and a benchmark garden before embedding R&D, repair, and standards workshops in existing edges.

AI Origin | campus-city shared courts



2021 vegetation signal

Prioritise existing ground floors and micro-renewal. Shared Eave, free study, and care waiting translate university knowledge into civic use.

Dazhongsi | multimodal arrival commons



2021 vegetation signal

Highest built-cover signal. Organise metro exits, bus, cycling, walking, frontages, and logistics as one reviewable public ground before making a landmark.

TWO FLAGSHIPS: DEPTH THROUGH SECTION AND INSTITUTION

Material / weather / operations / human fallback / exit carry equal weight

Qinghuayuan Shared Eave

- WHY** Let weather, waiting, learning, repair, and human help share one civic threshold; AI is not a display case.
- MODULE** Proposed 6m structural rhythm and 3m service/study submodules; independent foundations and heritage setback need specialist design.
- MATERIAL** Demountable steel/engineered timber frame, replaceable translucent weather panels, reclaimed-brick seats, open drainage, and snow-drip control.
- WEATHER** Summer shade and cross-ventilation; continuous dry movement in rain; sunny wind shelter in winter; clear route in snow.
- OPERATE** District/park coordination, professional inspections, community staffing; AI only provides explainable routing and staffed help always remains.
- STOP** Remove devices and retain minimum shelter/seating if heat, slip, noise, trees, heritage, fire, or maintenance tests fail.

Dazhongsi Four-Corner Commons

- WHY** Treat the station interface as one continuous public ground, not four styled corners or an underground retail shortcut.
- MODULE** Use the 300m station-integration control zone as a review frame; no redline, bridge, or underpass is concluded here.
- MATERIAL** High-friction permeable paving, demountable canopy, impact-resistant seats, tactile/high-contrast wayfinding, and protected logistics edge.
- WEATHER** Weather shelter at exits, queue shade in summer, sunny winter waiting, and continuous accessible movement in rain or snow.
- OPERATE** Joint duty by rail, bus, road, project, city management, and accessibility representatives; stagger arrival and delivery peaks.
- STOP** Use the project 3,190-person/h prediction as a stress test; reduce load if queues occupy the clear route or human takeover is late.

AI AGENT DISTINCTION: A REPLAYABLE ADVERSARIAL STUDIO

Not more concepts: persist inputs, failures, human judgments, and exit conditions



4 traces · 3 prompts · 5 tool/version records · input hashes · 4 human decisions

Explicitly not claimed: autonomous approval, representative consent, deterministic LLM prose, or official current conditions.

FROM ATTRACTIVE PROPOSAL TO ACCOUNTABLE PROJECT

Lead / permit / relative cost / O&M / phase / stop

PROJECT	LEAD	PERMIT	COST	O&M / STOP
Public observation + field verification	Planning organiser / survey + engagement teams	Data licence, privacy, and site-work permission	S	Annual data audit; STOP: Source, time, sample, or consent cannot be explained
Minimum works at five seams	Subdistrict + road/landscape/transport authorities	Road, tree, drainage, accessibility, and fire review	M	Subdistrict coordination + work orders; STOP: Requires an unreviewed bridge/tunnel or creates detour without alternative
Reversible Shared Eave pilot	Park operator + locality + design team	Heritage, structure, fire, trees, and noise	M	Park maintenance + community roster; STOP: Heritage, tree, night-quiet, or maintenance threshold fails
Four-Corner ground protocol	District coordination + rail/bus/road/project parties	Traffic impact, rail protection, fire, accessibility	M-L	Joint duty + timed logistics; STOP: Peak clear route, egress, or human takeover fails
Three AI urban sandboxes	Scenario host + data owner + third-party evaluator	Data protection, cybersecurity, ethics, equipment/site permission	S-M	Named human owner + public learning receipt; STOP: No exit, no human alternative, or energy/complaints exceed threshold
2031-2035 adaptation	To be set by formal planning and evaluation	Fifteenth Plan evaluation + later formal policy	unknown	No advance commitment; STOP: No formal policy, budget, owner, or public mandate

REGIONAL ROLE | JINGZHANG IS NOT AN ISOLATED TECH PARK

Four Beijing duties → Haidian's Two Zones and One Belt → AI urban testing and public-value verification

BEIJING

Global innovation centre · historic city · garden city · people's city

HAIDIAN

World-leading science park · Two Zones and One Belt · education-research-talent

JINGZHANG

Real-city testing, public-value verification, and explanation before scaling

AI URBAN TEST + PUBLIC-VALUE VERIFICATION CORRIDOR

Accessibility

Data minimisation

Energy

Climate

Maintenance

Inclusion

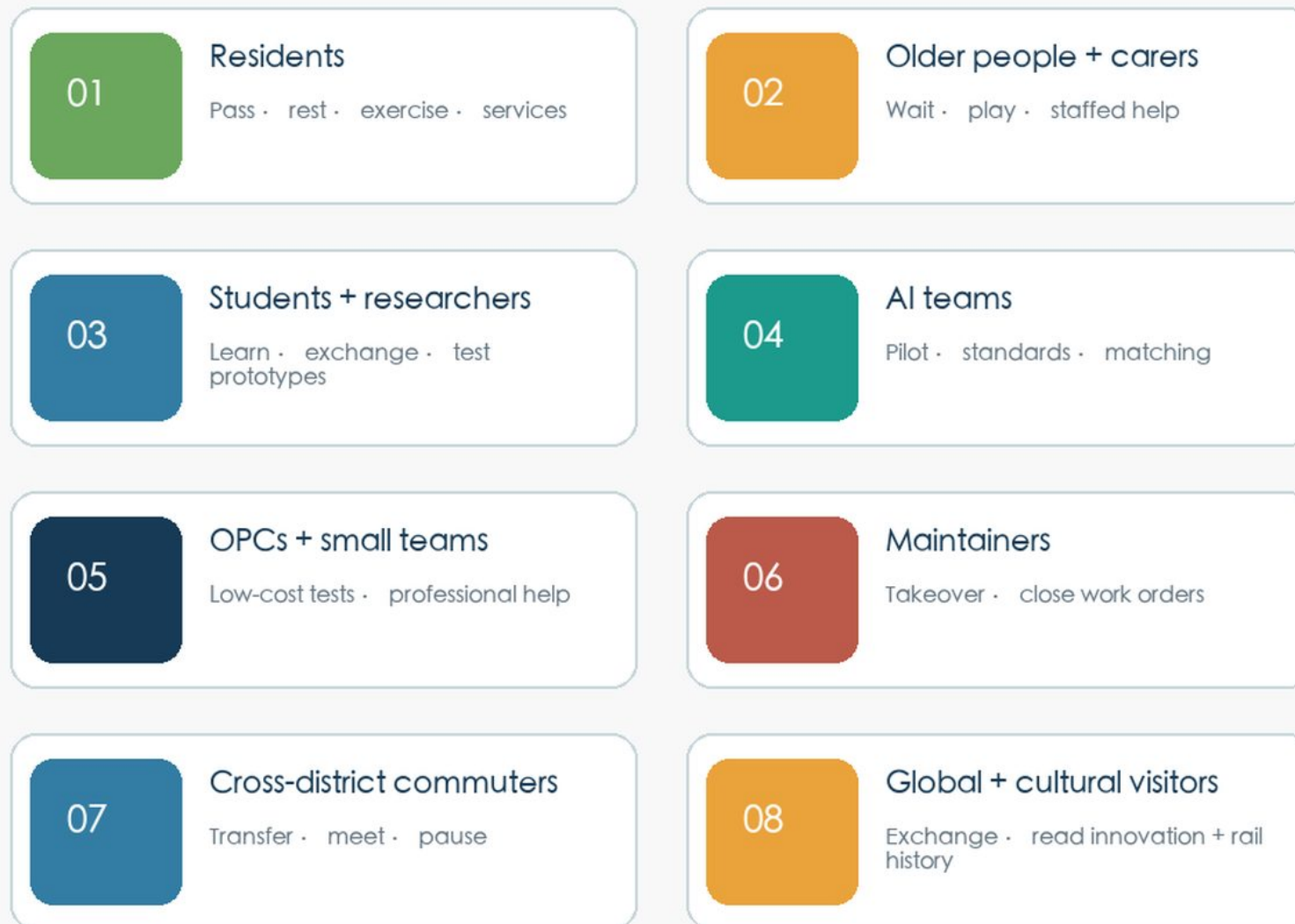
Pause + exit

Basis: adopted Beijing and Haidian 15th Five-Year Plans plus Beijing 2035 Master Plan; this is a design translation, not an official designation or approval.

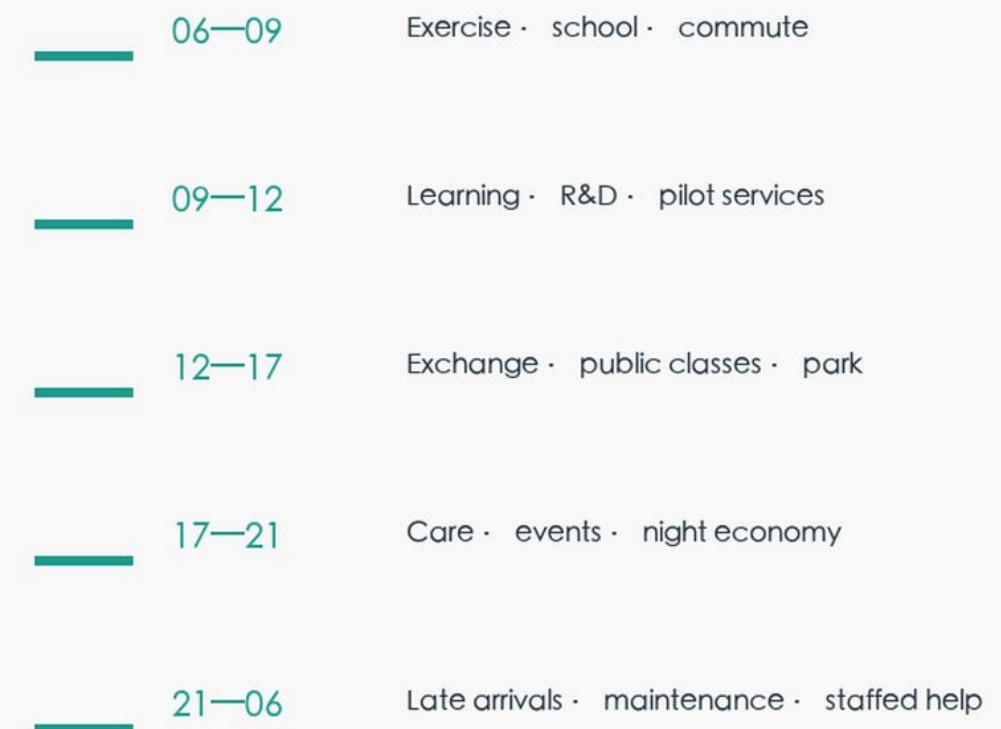
Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

PEOPLE + TIME | WHO COMES, AND WHY?

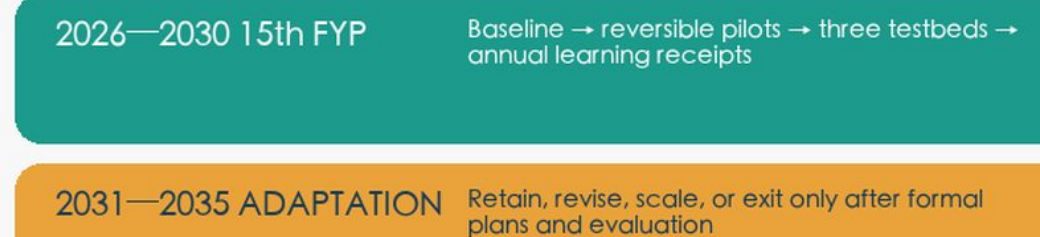
District statistics set context; eight groups are fieldwork hypotheses, not a concluded resident profile



A DAY OF FLOWS



PLANNING HORIZONS



Haidian context: 3.111m residents and 33% non-hukou residents in 2025; 18.5% were aged 60+ in the 2020 census. District context cannot replace site fieldwork.

Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

OVERALL FRAMEWORK | ONE SPINE + FIVE SEAMS

Three districts, two support wings, and ten work packages form a belt-wide public-infrastructure protocol



Civic-climate spine

Heritage · mobility · ecology · service

Five cross-city seams

Campus-park-river-station public links

Three differentiated districts

Full stack / origin / AI-native economy

Two support loops

Professional services + real-world tests

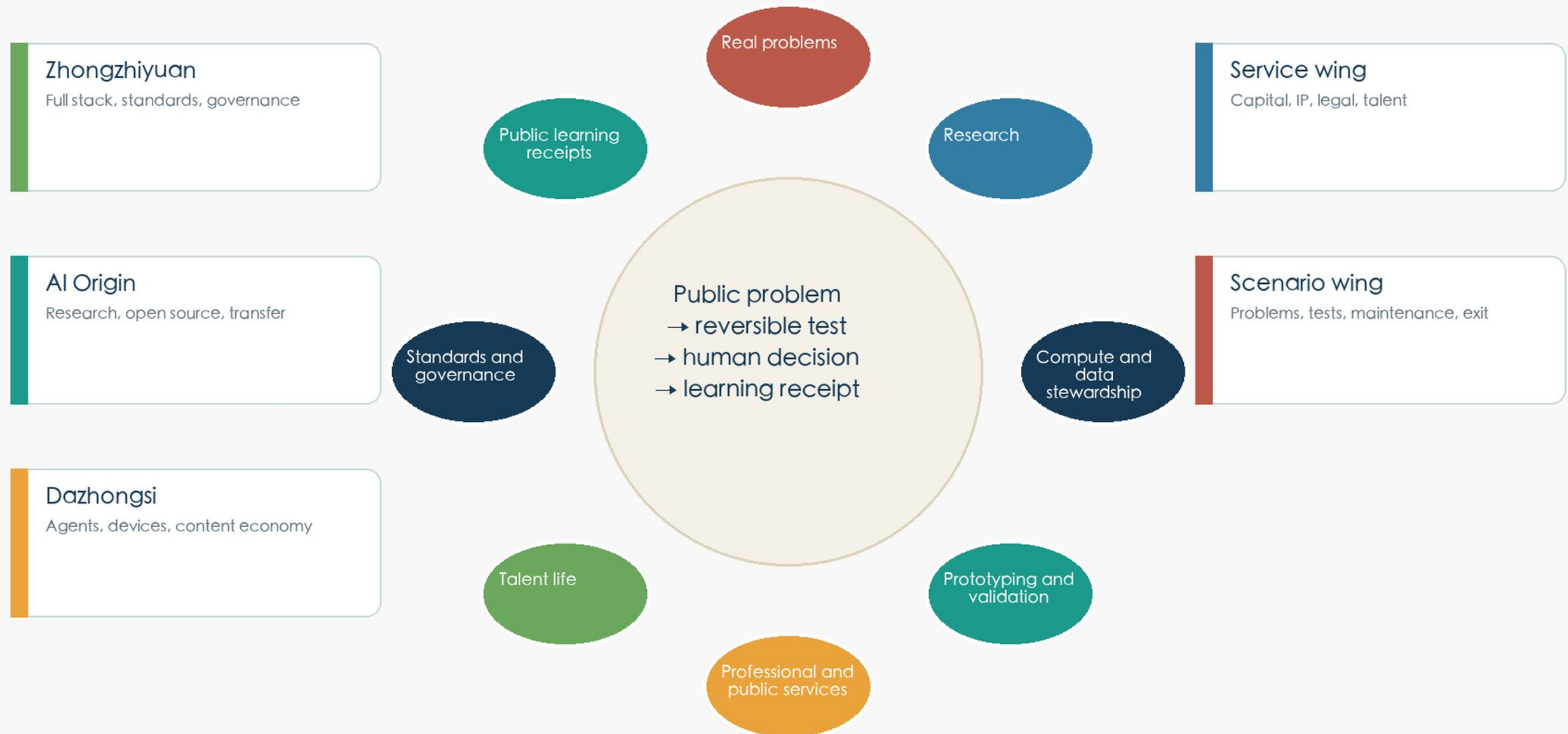
Ten renewal work packages

Data gate-pilot-design-public-value review

Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

INNOVATION ECOSYSTEM | EIGHT ELEMENTS FORM A LEARNING LOOP

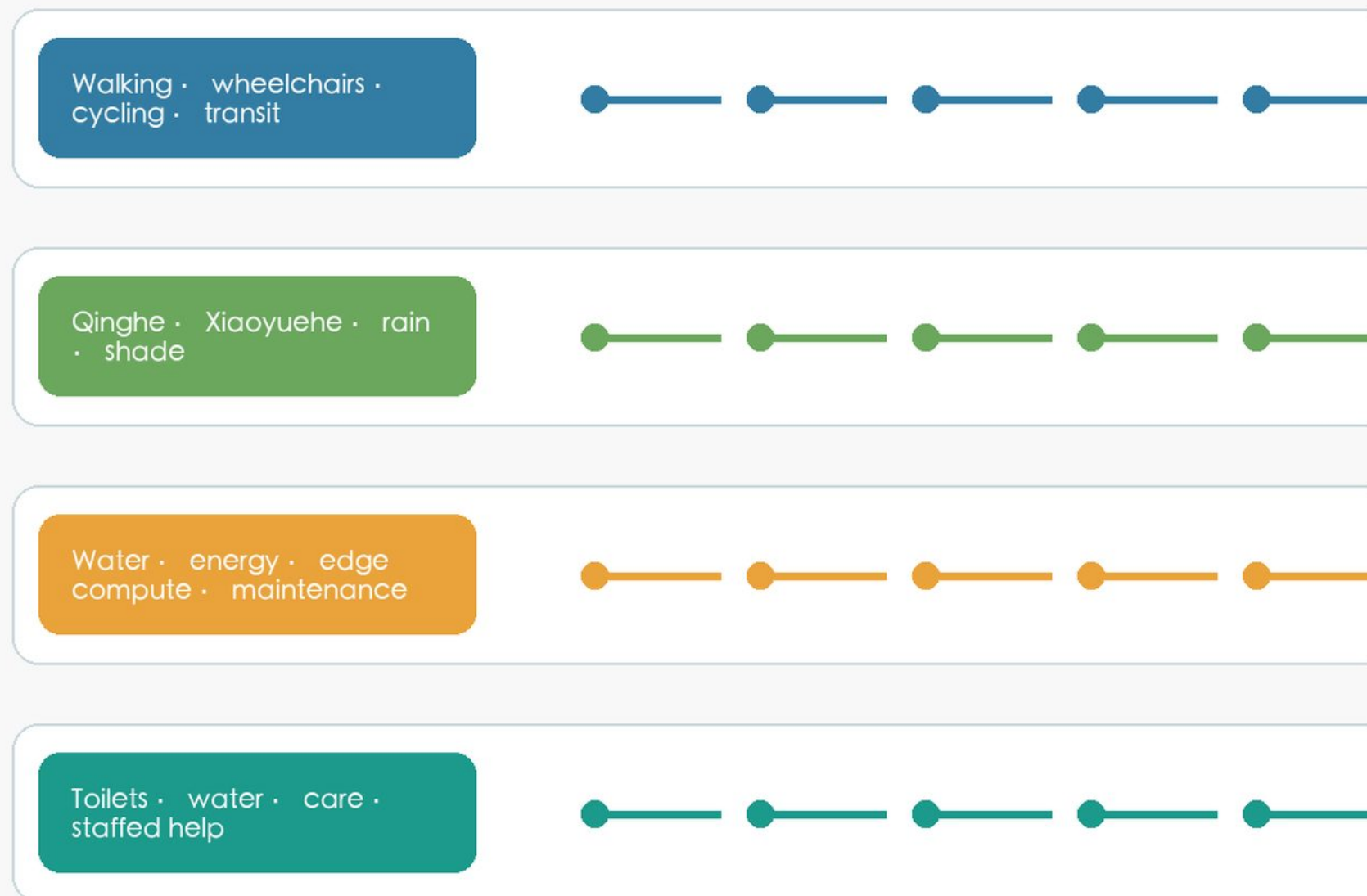
Space is not a container for company lists; it repeatedly connects problems, research, testing, services, and governance



Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

FOUR NETWORKS | ONE RESPONSIBILITY CHAIN

Mobility, landscape, utilities, and public service resolve access and maintenance before heavier engineering



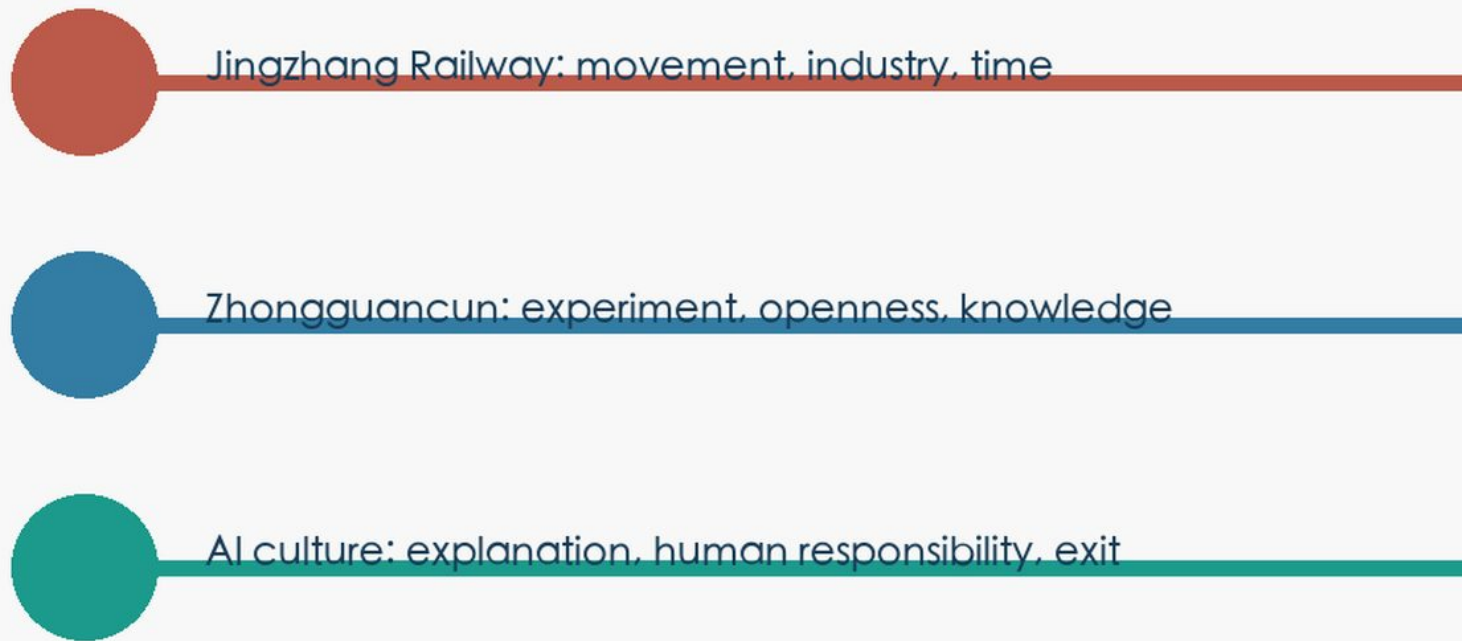
COMMON RESPONSIBILITY CHAIN



Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

CULTURE + OPERATIONS | REPEATED CIVIC PRACTICE

Railway time, Zhongguancun openness, and accountable AI culture form the Common Eave



IDENTITY GRAMMAR

Line: railway time and urban seams
Plane: shared shelter and free pause
Point: staffed help and contribution honour
Colour: navy, Qinghe green, signal gold; no corporate logo or cyber-neon

YEAR-ROUND OPERATING LOOP

SPRING | public problem call

SUMMER | reversible prototypes

AUTUMN | public review + global week

WINTER | maintenance + exit

Problem → prototype → review → continue / revise / pause / exit

Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

THREE KEY AREAS | THREE URBAN TYPES, NOT THREE COPIES OF ONE PARK

Each area has its own industry mechanism, spatial structure, civic lever, and delivery risk

ZHONGZHIYUAN | Garden innovation district

INDUSTRY MECHANISM

Full stack + standards governance

CIVIC LEVERS

Benchmark Garden
Open-stack commons
North gateway

DATA + PROFESSIONAL GATES

Fifth Ring access, Qinghe water, safety standards

AI ORIGIN | Near-campus innovation district

INDUSTRY MECHANISM

Research + incubation + transfer + life

CIVIC LEVERS

Shared Eave
Open transfer court
Near-campus life loop

DATA + PROFESSIONAL GATES

Heritage, campus edge, rent, existing users

DAZHONGSI | Urban innovation district

INDUSTRY MECHANISM

AI-native economy + global exchange

CIVIC LEVERS

Four-Corner Commons
South park foyer
Station-city service

DATA + PROFESSIONAL GATES

Four-quadrant traffic, transit, night operations

Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

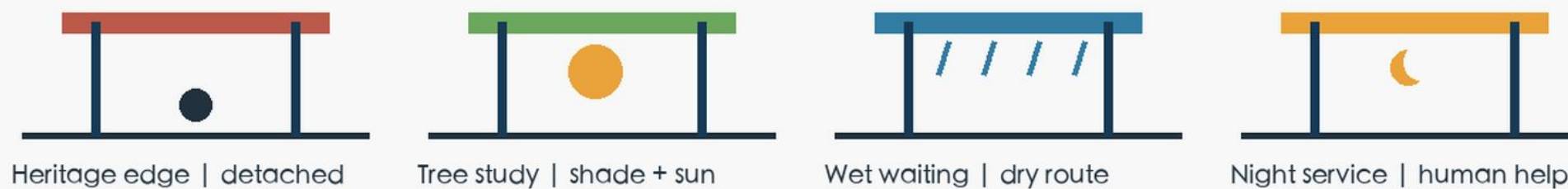
ARCHITECTURAL DEPTH | SHARED EAVE

A linear civic room: detached from heritage, seasonal, maintainable bay by bay



HERITAGE FABRIC

Review buffer: no invented statutory setback



FOUR GATES

01 Survey + heritage
Geometry, trees, fire

02 Reversible pilot
Furniture, signs, staff

03 Professional design
Architecture, structure, drainage,
access

04 Evidence-led replication
Scale only proven bays

Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

FOUR-SEASON CIVIC CLIMATE | SHARED EAVE

Architecture is not one roof; it is a set of usable seasonal conditions

WINTER | borrow sun



Sunny bays and wind-protected backs
No sealed glass shortcut

SPRING | receive rain



Roof water to outer rain garden
Dry route apart from drainage

SUMMER | borrow shade



Existing canopy first
Reversible shade without root harm

AUTUMN | open class



Open tables and static panels
Learning remains when tech stops

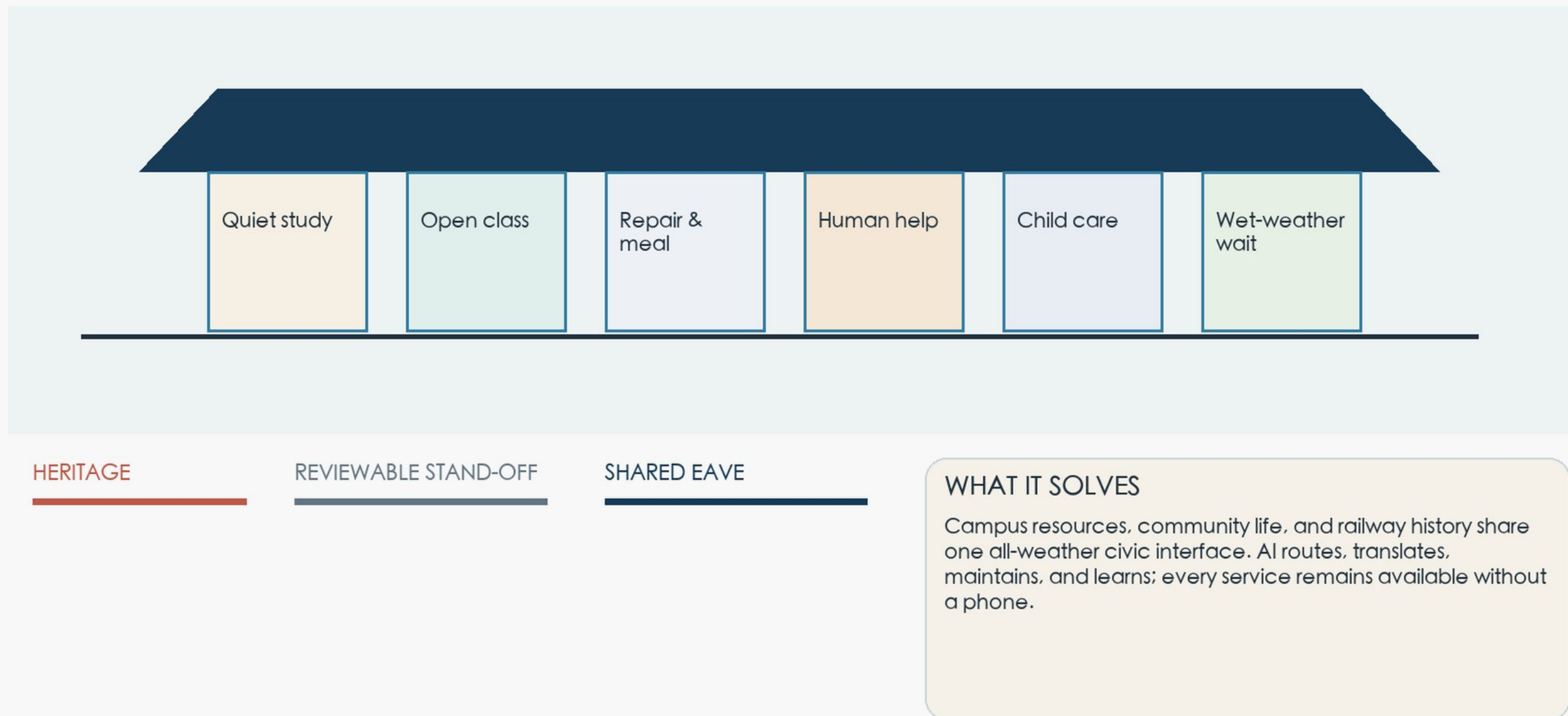
COMMON BASELINE

Continuous accessible ground · free pause · visible staffed help · static bilingual wayfinding · isolate one bay, keep others open ·
real comfort awaits professional modelling and field tests

Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

CIVIC PROTOTYPE 1 | QINGHUAYUAN SHARED EAVE

Detached from heritage, self-supporting, reversible; one eave holds a full day of civic life



Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

ATMOSPHERE | QINGHUAYUAN SHARED EAVE

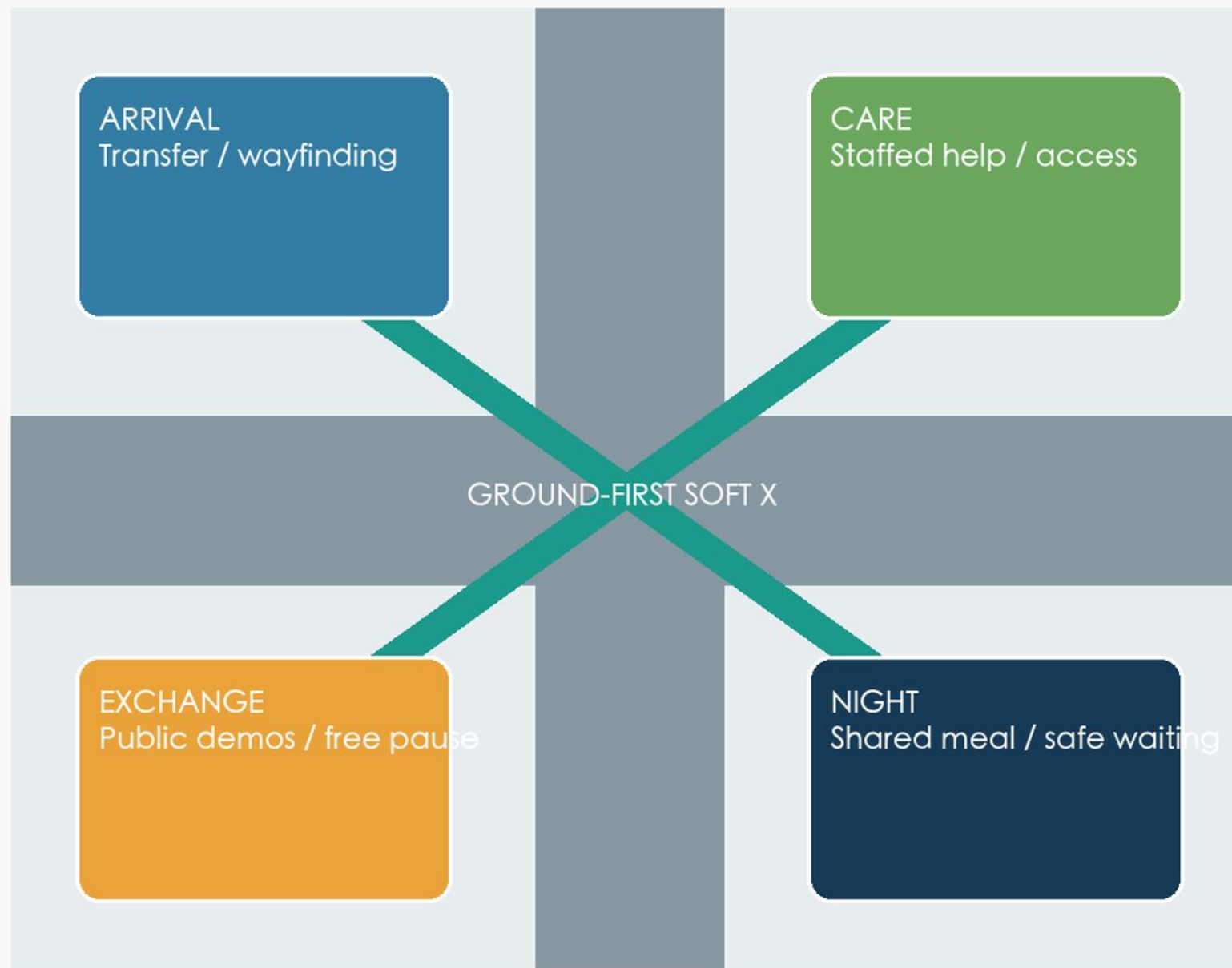
AI-generated concept atmosphere | Not a site photo, engineering location, or dimensional evidence



Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

GROUND CIVIC BUILDING | FOUR-CORNER COMMONS

No bridge first: organise arrival with four rooms, continuous ground, and time



24 HOURS, NOT ONE FIXED PLAN

06-10 Arrival priority

10-16 Public service

16-20 Shared curb

20-24 Staffed night

THREE REJECTIONS

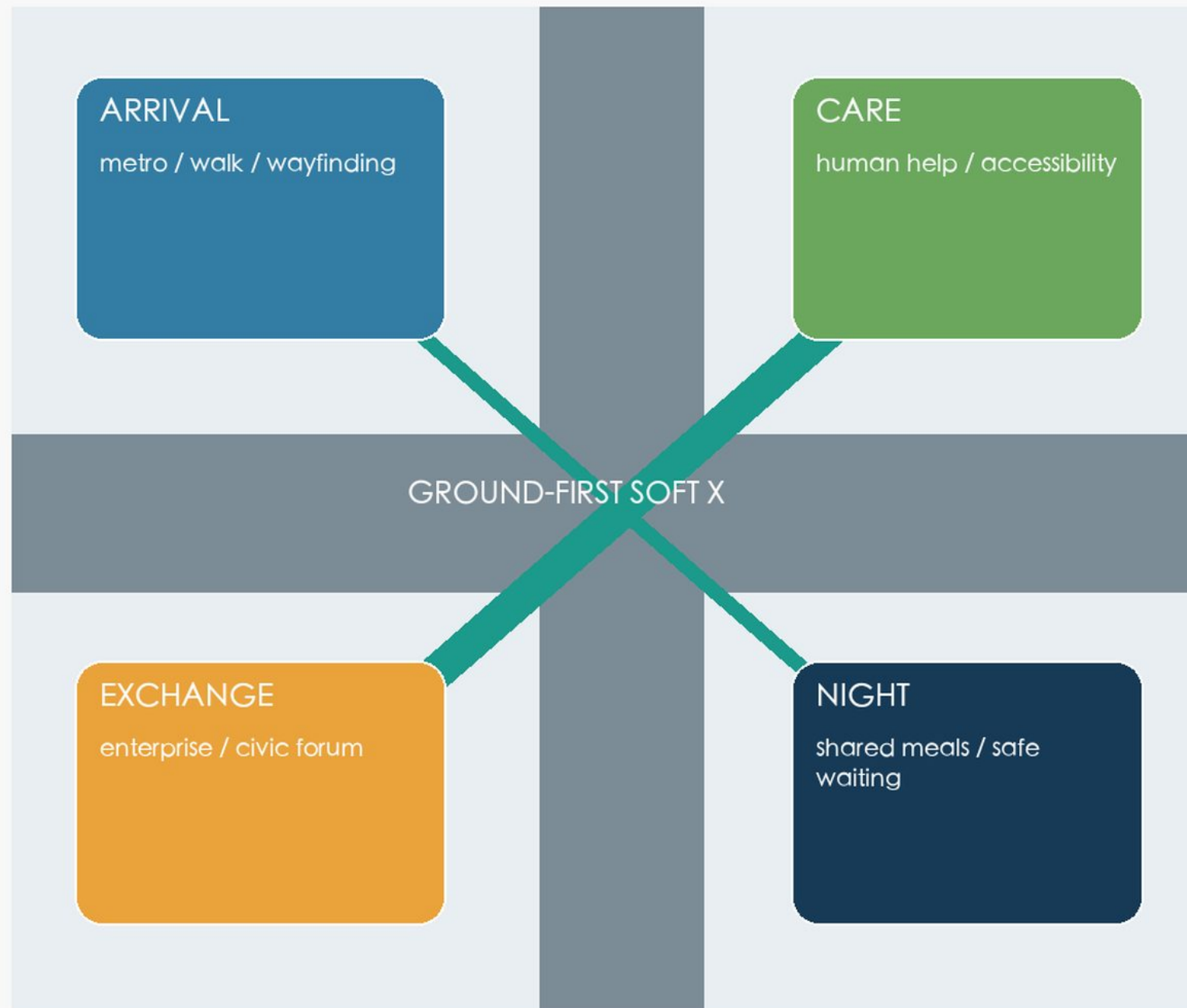
- × Commercial seats replace free pause
- × Agent changes real signals
- × Storm safety claimed without drainage evidence

Fallback: shrink, human takeover, pause, or keep unknown.

Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

PROTOTYPE 2 | FOUR-CORNER COMMONS

Do not start with a bridge: organise the ground, four corners, time, and responsibility first



SHARE TIME, NOT JUST SPACE

Commute: crossings and cycle arrival
Midday: public service and exchange
Evening: community and safe waiting
Event days: human takeover, automation paused

AGENT DOES FOUR THINGS

Forecast waits and conflicts
Propose time windows
Explain trade-offs
Record human decisions and learning

No face recognition, no autonomous signal change,
no replacement of transport or planning approval.

Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

ATMOSPHERE | DAZHONGSI FOUR-CORNER COMMONS

AI-generated concept atmosphere | Not a site photo, engineering location, or dimensional evidence



Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

TWELVE CIVIC JOURNEYS | NOT TWELVE INVENTED OPINIONS

Agents use journeys to find problems; public judgment still needs consented walks, interviews, and co-design

SHARED EAVE

Older neighbour Winter morning | no smartphone

Sunny seat, toilet wayfinding, staffed help

Wheelchair user Summer shower | wet ground

Continuous dry route, turning space, step-free service

Child and carer Evening | after class

Observable play, water, quiet waiting

Young researcher Afternoon | open class

Bookable table, power, non-commercial study

International visitor Network outage | first visit

Static bilingual wayfinding, human explanation, railway story

Maintenance worker Night | local component failure

Dry replacement, isolate one bay, keep the rest open

FOUR-CORNER COMMONS

Wheelchair transfer user AM peak | four-quadrant transfer

Continuous ground, less detour, rest points

Cycle commuter AM peak | parking conflict

Clear separation, short stay, no crossing blockage

Carer with child Midday | free pause

Free seating, visible toilet route, traffic buffer

Night-shift worker Late night | solo arrival

Staffed help, restrained lighting, visible assistance

Delivery and service worker Evening | system offline

Manual schedule, legible curb, event pause option

Public service worker Storm | demand surge

Dry waiting, drainage observation, human takeover

Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

EVIDENCE CHAIN | WHAT IS KNOWN, CONCEPTUAL, OR PENDING

Machine gates validate structure and recomputation; professional data, public judgment, and field performance remain pending

20

Planning units

5

Cross-city seams

10

Renewal packages

12

Building interfaces

9

Mobility links

11

Green components

VERIFIED RELATIONSHIPS

- Twenty units cover the provisional boundary without gaps or overlaps
- Areas and ratios recompute from the same GeoJSON
- Twelve Agent tasks and three negative samples are replayable
- Chinese and English chapters, figures, HTML, and PDFs align independently

PASS = ready for content review; it is not excellence, selection, implementation, field safety, or endorsement.

Concept proposal | Provisional spatial constraint, not an official boundary | 2026-08-10

PENDING OFFICIAL DATA

- Official geometry and property rights
- Surveyed buildings and building-level action
- FAR, height, density, and setbacks
- Road lines, rail, flows, and parking
- Utilities, drainage, energy, fire, and flood
- Heritage limits, trees, and ecology
- Capital, operating cost, and responsible bodies
- Consented public engagement and acceptance